

FIELD REFERENCE

The Videographer's Cheat-Sheet Pack

Grab-and-Go Cards for Frame Rates, Audio & Export

Story Mode Guy

storymodeguy.com

Reader's Edition · 2026

\$17 value

What's Inside

Grab-and-Go Cards for Frame Rates, Audio & Export. Written for new shooters who want a printable field reference.

01 How to Use This Pack

02 Frame Rate & Shutter Angle

03 Resolution, Bitrate & Codec

04 Picture Profile Quick-Set

05 Exposure Tools

06 Audio Levels & Mic Checklist

07 Lighting Quick-Reference

08 The B-roll & 5-Shot Shot-List

09 Camera Movement Cheat Card

10 Export Settings & Pre-Shoot Checklist

START HERE

How to Use This Pack

This is not a manual you read once and forget. It's a deck of field cards — each one dense, self-contained, and built to live in your bag. Print them, keep them close, and pull the right card the moment you're set up, the light is going, and you're blanking on what to dial in.

Every card that follows is a standalone reference. There's a card for frame rates and shutter, a card for codecs and bitrate, cards for picture profiles, exposure tools, audio, lighting, movement, shot lists, and one packed with export recipes. Nothing depends on anything else. Grab the one you need and go.

HOW TO GET THE MOST FROM THIS PACK

Print each card, keep them in your camera bag, and reach for the frame-rate card and the audio checklist first — those two save more footage than anything else here.

Three ways to use these cards



Print & keep

Each **card starts on its own page**, sized to trim down and slip into a pocket. Laminate them so they survive rain, sweat, and the bottom of your bag.



Keep on your phone

Save the PDF to your phone. When you're on location and unsure, the shutter-angle math and export recipes are two taps away.



Learn by repetition

Shoot the settings as written, then adjust. After a few projects the numbers live in your hands and you won't need the card at all.



Works with any camera

Every setting here applies to mirrorless bodies, cinema cameras, DSLRs in video mode, and most phones in Pro/Cinematic mode. Where a control has different names by brand, both are given (for example, "Log (S-Log / V-Log / C-Log)"). Treat every number as a *starting point*, not a law — check your footage and nudge from there.



The one habit that ties it all together

Before you hit record, run a three-second check: **frame rate** (does it match the project?), **shutter** (is it double the frame rate?), and **audio** (are the meters actually moving?). These cards exist to make that check automatic.

CHAPTER 2

Frame Rate & Shutter Angle

Frame rate is how many still images you capture per second; shutter speed is how long each one is exposed. Get the relationship between them right and your motion looks natural — like film. Get it wrong and footage looks either stuttery or oddly video-game smooth.

The 180° rule

For natural motion blur, set your **shutter speed to roughly double your frame rate**. That's what a 180-degree shutter angle means. Shoot 24 fps, use 1/48s (round to 1/50s). Shoot 30 fps, use 1/60s. Simple as that.

Frame rate	Shutter (180° rule)	Shutter angle	What it's for
24 fps	1/48s (use 1/50s)	180°	The cinematic look — films, narrative, most YouTube
25 fps	1/50s	180°	PAL regions (Europe), broadcast, film look
30 fps	1/60s	180°	Web video, social, news, live-feel content
60 fps	1/120s	180°	Smooth motion, sports, half-speed slow-mo
120 fps	1/240s	180°	Dramatic slow-mo, water, action detail



Angle vs. speed

Cinema cameras let you set a **shutter angle** directly (180° is the default). Most hybrid cameras only show shutter *speed*. They describe the same thing — angle just adapts automatically when you change frame rate, so you never have to recompute.

Slow-motion math

Slow motion is simple: **shoot at a high frame rate, play it back at a normal one**. The playback rate divided into the capture rate is your slowdown factor.

How slow will it go?

FIELD-TESTED

60 fps → 24 fps = 2.5× slower

60 fps → 30 fps = 2×

120 fps → 24 fps = 5×

240 fps → 24 fps = 10×

Always match your slow-mo clip's project frame rate. Shoot 120, drop it on a 24 fps timeline, and it plays back five times slower automatically.



High frame rates eat light

At 120 fps your shutter is 1/240s — half as much light per frame as 60 fps. Slow-mo needs bright scenes or more ISO. Plan to shoot it outdoors or add light.



Pick one frame rate per project

Mixing 24 and 30 fps in the same edit is a headache. Choose your base rate up front (24 for cinematic, 30 for web) and only break from it for intentional slow-mo clips.

CHAPTER 3

Resolution, Bitrate & Codec

Resolution is how many pixels; bitrate is how much data per second the camera spends describing them; codec is the method used to compress that data. Together they decide how sharp your footage is, how much it holds up in the edit, and how fast it fills your card.

Resolution — how many pixels

Resolution	Pixels	Use for
1080p (Full HD)	1920 × 1080	Fast turnaround, smaller files, most social delivery
4K (UHD)	3840 × 2160	The modern default — crop room, future-proof, punch-ins
6K / 8K	6144+ wide	Heavy reframing, VFX, big-screen delivery



Shoot 4K, deliver 1080p

Even for a 1080p final, shooting 4K gives you room to reframe, punch in for a fake "second angle," and stabilize without losing quality. Downscaled 4K also looks noticeably sharper than native 1080p.

Codec & bitrate — how the data is stored

Codec	Type	Rough bitrate	Use when...
H.264	Compressed (8-bit)	50–100 Mbps	Everyday shooting, small files, easy playback
H.265 / HEVC	Compressed (10-bit)	50–150 Mbps	Same quality as H.264 at smaller size; more color depth
ProRes / ProRes HQ	Lightly compressed	440–880 Mbps	Serious grading, client work, minimal quality loss
RAW / ProRes RAW	Near-uncompressed	1000+ Mbps	Maximum flexibility, VFX, high-end delivery



Bit depth matters more than you think

8-bit holds 16.7 million colors; **10-bit** holds over a billion. If you plan to grade — especially Log footage — shoot 10-bit so skies and skin tones don't band into ugly stripes.



Match your card to your bitrate

ProRes and high-bitrate recording demand fast cards. Check the card's **sustained write speed** (a V60 or V90 SD card, or CFexpress for the highest rates). A slow card causes dropped frames and stopped recordings mid-take.

CHAPTER 4

Picture Profile Quick-Set

A picture profile decides how much contrast and color the camera bakes into the file. Pick the one that matches how much you plan to edit — shoot flat when you'll grade, shoot standard when you won't.



Standard

Contrast & color baked in. Looks good straight out of camera. Least editing room. Best for run-and-gun and same-day delivery.



Flat / Neutral

Reduced contrast. Keeps more shadow and highlight detail while still looking usable raw. A gentle middle ground for light grading.



Log

Maximum dynamic range, gray & flat. Must be graded. The most highlight and shadow latitude — for when the look is made in post.

When to shoot each

Profile	Names by brand	Shoot it when...
Standard	Standard / Natural / Rec.709	Fast delivery, no time to grade, run-and-gun events
Flat	Neutral / Cine / HLG	You want some grading room but a usable image on-set
Log	S-Log / V-Log / C-Log / Log3	You'll grade every clip and want maximum latitude

EXPOSE LOG TO THE RIGHT

Log footage is designed to be shot slightly bright — often 1 to 1.5 stops over "correct." A brighter Log file has cleaner shadows once you pull it back down in the grade. Underexposed Log gets noisy fast.



Log has a minimum ISO

Log profiles have a high **base ISO** — often 640, 800, or higher — where they're cleanest. Going below it isn't "cleaner," it clips highlights. Learn your camera's Log base ISO and treat it as the floor.



Don't grade blind

Load a **LUT** to your monitor or viewfinder so you preview a corrected image while shooting Log. The recording stays flat; only your preview is transformed, so you can judge exposure and color accurately.



New to grading? Start with Flat

Log is powerful but unforgiving. If you're still learning, a flat/neutral profile gives you 80% of the flexibility with far less chance of a noisy, muddy, or clipped clip you can't rescue.

CHAPTER 5

Exposure Tools

In video you can't just change the shutter to fix brightness — the 180° rule locks your shutter to your frame rate. So you control exposure with three other tools: aperture, ISO, and ND filters. This card covers the filters and the on-screen tools that tell you the truth about your exposure.

ND filters — sunglasses for your lens

An ND (neutral density) filter darkens the image without changing color, so you can keep your shutter at the 180° rule *and* your aperture wide open in bright light. Without one, daylight forces you to a fast shutter that ruins your motion blur.

ND strength	Stops cut	Use in...
ND8	3 stops	Overcast, shade, golden hour
ND16	4 stops	Bright but not harsh daylight
ND64	6 stops	Full sun, open sky
ND1000	10 stops	Very bright sun at a wide aperture
Variable ND	2–8 stops (adjustable)	Run-and-gun — dial it as the light changes



One variable ND does most jobs

A quality **variable ND** (2–5 or 3–8 stop) covers most daylight situations from a single filter. Buy one that fits your largest lens and use step-up rings for the rest. Cheap ones add a color cast or an "X" pattern at the extremes.

Zebras & false color — judge exposure, not guesswork

Your screen's brightness lies; the sun on it lies harder. Use these tools instead of trusting your eyes.



Zebras

Diagonal stripes appear on areas hitting a brightness threshold you set. **Set to ~100%** to see clipping (blown highlights). **Set to ~70%** to expose skin tones consistently.



False color

Paints the image by exposure value — a full map from clipped to crushed. **Green/gray = mid-tones, red = clipping, purple/blue = crushed shadows.** The most precise tool once you learn the palette.

Skin-tone exposure with zebras

STARTING POINT

Zebra level ~70%

Target zebras on cheek

Log expose +1 stop

Set zebras to 70% and expose so they just tickle the bright side of your subject's face. Consistent skin exposure across every shot makes editing painless.



Highlights don't come back

Blown-out (clipped) highlights hold no detail — no grade recovers them. When in doubt, protect the highlights and lift the shadows later. A histogram or false-color pushed hard right is your warning.

CHAPTER 6

Audio Levels & Mic Checklist

Viewers forgive rough footage; they switch off bad audio in seconds. Good sound is half of "professional." This card gets your levels safe and your mic choice right.

Recording levels — aim for the safe zone

Level (dBFS)	Meaning	Action
0 dBFS	Absolute ceiling — clipping	Never touch it; clipped audio is unfixable
-6 dBFS	Loudest safe peaks	Let occasional peaks reach here, no higher
-12 dBFS	Ideal average for speech	Aim your normal talking level here
-20 dBFS	Quiet / too low	Raise gain — recording this low adds noise when boosted

TARGET -12, PEAK -6

Set your gain so normal speech sits around -12 dBFS and the loudest moments never pass -6. Headroom is safety — a laugh or a shout should never clip.

Delivery loudness — the platform standard

Your *recording* level and your *final export* loudness are different jobs. Streaming platforms normalize to a loudness target measured in LUFS. Mix to it so your video isn't jarringly loud or quiet next to others.

Platform	Loudness target
YouTube	-14 LUFS
Instagram / TikTok	-14 LUFS
Podcast / spoken	-16 LUFS
Broadcast TV	-23 LUFS (EBU R128)

Mic choice — right tool, right distance

Mic type	Best for	Watch out
Shotgun (on-cam / boom)	Interviews, run-and-gun, dialogue	Keep it close — 60cm beats 2m every time
Lavalier (clip-on)	Talking heads, moving subjects, presenters	Clothing rustle; hide it well
Wireless lav system	Distance, multiple people, freedom	Interference; always monitor
Handheld	Vox pops, stand-ups, music	Handling noise; needs a grip technique



Distance beats price

A cheap mic 30cm from the mouth sounds better than an expensive one two meters away. The single biggest audio upgrade is **getting the mic closer to the source**.

Field-audio checklist

- ☒ Headphones plugged in — you can't fix what you can't hear
- ☒ Levels set: speech peaking around -12, never past -6
- ☒ Windscreen / dead-cat on outdoors, always
- ☒ Fresh batteries in mics and transmitters; spares in the bag
- ☒ Record a few seconds of room tone at every location
- ☒ Turn off buzzing fridges, A/C, and fluorescent hums where you can
- ☒ Monitor the whole take — don't just set and pray

CHAPTER 7

Lighting Quick-Reference

You don't need a truck of gear to light well — you need to understand two qualities and three jobs. Master those and one light plus a window will carry you far.

Soft vs. hard — it's all about size

Soft light

From a **large source relative to the subject** — a big softbox, a bounced light, an overcast sky. Gentle, wrapping shadows. Flattering for faces and skin.

Hard light

From a **small or distant source** — a bare bulb, direct sun. Crisp, defined shadows. Dramatic, edgy, high-contrast. Great for mood.



Bigger and closer = softer

To soften any light, make its *apparent* size bigger: move it closer, bounce it off a wall or ceiling, or shoot it through diffusion. A window with a sheer curtain is one of the best soft sources on earth.

Three-point lighting — the roles

Key light

Your **main light**. Placed ~45° to one side of the subject. Sets the exposure and the overall look.

Fill light

Softens the shadows the key creates, on the opposite side. Weaker than the key — or just a bounce card.

Back / rim light

Behind the subject, edging their hair and shoulders to **separate them from the background** and add depth.

Color temperature — mixing warm and cool

Source	Color temp	Look
Tungsten / warm bulb	3200K	Warm, orange, cozy
Golden hour sun	~3500K	Warm glow
Daylight / flash	5600K	Neutral, clean white
Overcast / shade	6500K+	Cool, blue



Match your light to the room

Mixing a 5600K daylight LED with 3200K tungsten room lamps gives you one orange side and one blue side of a face. Either **gel your light to match** the room or kill the room lights and control it all yourself.

One-light interview setup

STARTING POINT

Light 45° to side

Height just above eye

Quality softbox / bounce

Fill white bounce card

One soft key at 45°, a bounce card on the shadow side, and a window or lamp behind for separation. That's a clean, professional interview with a single fixture.

CHAPTER 8

The B-roll & 5-Shot Shot-List

B-roll is the supporting footage that makes an edit breathe — the details, the wides, the hands at work. The fastest way to cover any subject or scene is the 5-shot method: five deliberate angles that always cut together.

The 5-shot method

Shot	What it captures	Why it works
1. Wide (establishing)	The whole scene / location	Tells the viewer where they are
2. Medium	The subject and their action	The main "who and what" shot
3. Tight (close-up)	The face or the focus of action	Emotion and detail; the money shot
4. Detail (extreme close)	Hands, tools, textures, product	The cutaway that hides edits
5. Reverse / over-shoulder	The opposite angle or POV	Gives the sequence a second dimension

FIVE SHOTS CUT INTO A SCENE

Get all five of any moment — a person cooking, a mechanic working, a street performer — and you can edit a complete, dynamic sequence. Coverage beats camera gear every single time.

Reusable B-roll shot-list

- ☒ Wide establishing shot of the location
- ☒ Medium shot of the subject doing the action
- ☒ Tight close-up of the face / reaction
- ☒ Extreme detail: hands, tools, product, texture
- ☒ Reverse angle or over-the-shoulder / POV
- ☒ A slow, static "beauty" shot for openers & transitions
- ☒ One deliberate camera-movement shot (push, reveal, or pan)



Entrances and exits — someone walking into and out of frame



Hold every shot for ten seconds

Count to ten on each B-roll clip, and don't cut the camera the instant the action ends. Short clips are useless in the edit; a generous hold gives your editor room to breathe and time the cut.

CHAPTER 9

Camera Movement Cheat Card

Movement adds energy and meaning — but only when it's motivated. Every move should have a reason: to reveal, to follow, to draw the eye. This card covers the core moves and when each one earns its place.

Pan

Pivot the camera **horizontally** on a fixed point. Follow a subject or scan a landscape. Keep it slow and smooth.

Tilt

Pivot **vertically**. Reveal height — a tall building, a full figure from feet to face.

Push in / Pull out

Move the whole camera **toward or away** from the subject. Push in to build tension or intimacy; pull out to reveal context.

Tracking / Dolly

Move **alongside or with** a moving subject. Keeps them steady in frame while the world slides by — dynamic and immersive.

Reveal

Move past a foreground object to **uncover** the subject behind it. Adds depth and a little drama to any entrance.

Whip pan

A fast, blurred pan used as a **transition**. Match the direction between two clips for a seamless energetic cut.

Motivation over motion

A locked-off static shot is often stronger than a wandering handheld one. Move the camera when the story asks for it — to follow, reveal, or intensify — not just because you can.

Start and end on a hold

Begin every moving shot with a second or two of stillness, execute the move slowly, then hold still at the end. Those static bookends give your editor clean in and out points and make any move look intentional.

Stabilize on a budget

No gimbal? Tuck your elbows in, use the "ninja walk" (bent knees, heel-to-toe), and let stabilization help. For push-ins, a slider beats walking.

CHAPTER 10

Export Settings & Pre-Shoot Checklist

The last mile. Nail your export and your footage looks as good online as it did on the card. These recipes are safe, proven starting points for the three deliveries you'll make most.

YouTube (1080p / 4K)

EXPORT RECIPE

Format H.264 / MP4

Resolution 3840×2160 or 1920×1080

Bitrate 1080p: 16 Mbps · 4K: 45 Mbps

Audio AAC 320 kbps

Match the project frame rate. YouTube re-compresses everything, so exporting a touch high (or using their higher-quality specs) preserves detail.

Instagram / TikTok (vertical)

EXPORT RECIPE

Format H.264 / MP4

Resolution 1080×1920 (9:16)

Bitrate 10-15 Mbps

Frame rate 30 fps

Shoot or reframe for vertical 9:16. Keep key action away from the edges where the interface covers the frame. Loudness -14 LUFS.

Client delivery (master)

EXPORT RECIPE

Format ProRes 422 HQ (or high H.264)

Resolution native project size

Bitrate near-lossless / 100+ Mbps

Audio 48kHz, -16 to -14 LUFS

Deliver a high-quality master the client can re-use, plus a compressed MP4 for easy review. Always keep your own archived copy.



Frame rate must match end to end

Shoot 24, edit on a 24 fps timeline, export 24. Mismatched frame rates cause stuttery, judder-y playback that no bitrate fixes. Set your sequence to your footage before you cut a single clip.

Final pack-and-shoot checklist

- ☒ Batteries charged — camera, mics, monitor, gimbal — plus spares
- ☒ Cards formatted, fast enough for your codec, and enough capacity
- ☒ Frame rate and shutter set (shutter = 2× frame rate)
- ☒ Picture profile chosen; LUT loaded to monitor if shooting Log
- ☒ ND filter in the bag (or on the lens) for daylight
- ☒ Zebras / false color enabled to nail exposure
- ☒ Mic connected, levels tested, headphones packed
- ☒ White balance set to the dominant light
- ☒ Lens clean; lens cloth in a pocket
- ☒ Shot-list printed — know your five shots before you arrive



Keep this card closest

Of everything in this pack, laminate this export card and the audio checklist first. Between "how do I export this?" and "why does my sound clip?", these two solve the problems that most often sink a finished video.

REMEMBER

You don't have to memorize videography — you have to carry the right card, check your frame rate and your levels, and roll, and these cards make sure you always can.

Story Mode Guy

Guides for people who treat everything like it has a plot. · storymodeguy.com